NTNU | Norwegian University of Science and Technology

Exercise 1

Requirement Engineering

TDT4242 Advanced Software Engineering

Group 3
Raffaella Corcione
Ruggero Facchini
Birgitte Louise Vik Thoresen

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Chapter 1

Task 1.1. Requirements Elicitation

1.1 Interview Transcript iRE Agent

Can you provide a general description of the application you would like to create?

The application we aim to create is the AI Guidebook for Students, designed to support academic work by helping students use generative AI tools responsibly. It will provide features such as AI usage logging, assignment-level AI declarations, ethical guidelines, and automated compliance checks with institutional policies. The system will also include a personal dashboard for visualizing usage patterns, ensuring transparency and privacy while promoting academic integrity.

Can you describe specific, measurable ways you'd want to track your AI usage that would be helping students use generative AI tools responsibly?

To track AI usage effectively, the system shall implement several measurable methods. First, it should log the frequency of AI tool usage per assignment, allowing students to see how often they rely on AI assistance. Second, it shall require students to declare the specific AI tools used for each assignment, promoting transparency. Third, the system can provide visual analytics on usage patterns over time, highlighting trends in AI dependency. Lastly, automated compliance checks against institutional guidelines should be included to ensure responsible usage aligned with academic integrity standards. Each of these elements can be quantified and analyzed to foster responsible AI tool use.

How would the assignment AI declaration be performed?

The assignment AI declaration shall be performed through an integrated form within the application. Students will be prompted to complete this form upon submission of each assignment, specifying the AI tools utilized. The system shall require the following fields: assignment title, AI tools used, purpose of AI assistance, and any generated content. This structured approach facilitates transparency and ensures compliance with academic integrity policies. Additionally, the declaration process should be straightforward to encourage consistent usage by students.

Should the AI guidelines compliance checks be done periodically? If relevant, what would be the frequency of these checks?

The system shall implement periodic compliance checks to ensure adherence to AI guidelines. The frequency of these checks could be set to occur at key milestones, such as after assignment submissions or at the end of each academic term, depending on institutional policies. It's essential to balance the frequency with the need for timely feedback and support for students in maintaining academic integrity.

What do you mean by "balancing the frequency" of the checks?

Balancing the frequency refers to finding an optimal schedule for compliance checks that provides timely feedback to students without overwhelming them. Too frequent checks may lead to unnecessary stress or administrative burden, while infrequent checks could result in missed opportunities for students to correct their AI usage and maintain academic integrity. It's crucial to establish a frequency that supports effective learning while ensuring compliance with guidelines.

Should these check also evaluate sustainability in the use of AI?

The system shall consider including sustainability evaluations as part of the compliance checks for AI usage. Evaluating sustainability would ensure that students are not only using AI responsibly but also being mindful of the environmental impact associated with these technologies. This could involve assessing the frequency and duration of AI tool usage and encouraging efficient practices in alignment with broader institutional sustainability goals.

Describe a typical workflow of a student using the application. How would the user interact?

A typical workflow for a student using the application may include the following steps:

1. Login: The student logs into the application and accesses their personal dashboard.
2. AI Tool Selection: When starting an assignment, the student selects from a list of AI tools they plan to use.
3. Assignment Declaration: The student completes an AI declaration by providing details about the tools used and a brief description of their contributions to the assignment.
4. Reflection: The system prompts the student to reflect on their AI usage regarding academic integrity, providing questions to consider.
5. Submission: After completing the assignment, the student submits it along with the AI declaration, which is logged for compliance checks.
6. Feedback and Compliance Check: The system periodically evaluates the AI usage in relation to guidelines and provides feedback, allowing the student to adjust their practices as necessary.

7. Review and Adjust: The student can review their usage patterns over time on their dashboard, making adjustments based on feedback and sustainability prompts.

This workflow ensures that students engage with the application in a structured manner while promoting responsible AI usage and compliance with academic integrity policies.

What details should the student provide during the assignment declaration steps?

During the assignment declaration steps, the student shall provide the following details:

1. AI Tools Used: A list of specific AI tools utilized during the assignment.
2. Description of Usage: A brief explanation of how each tool contributed to their work, including tasks supported by the AI.
3. Reflection on Academic Integrity: Insights on how their AI usage aligns with institutional academic integrity policies, addressing any potential ethical concerns.
4. Duration of Use: An estimate of the time spent using each AI tool during the assignment process.
5. Sustainability Considerations: Any efforts made to use AI tools efficiently and sustainably, if applicable.

These details will help ensure transparent and responsible use of AI in their academic work.

Can you make an example of "efficient and sustainable" AI usage?

An example of "efficient and sustainable" AI usage could be a student using an AI writing assistant to draft an essay. Instead of relying solely on the AI for the entire content, the student uses it to generate outlines and key points, then integrates these suggestions with their own research and writing. This approach minimizes excessive reliance on AI, reduces computational resources, and promotes critical thinking. Additionally, the student could limit AI usage to specific phases of the writing process, such as brainstorming and editing, thereby optimizing the time spent with the tool while being mindful of its environmental impact.

What tools should the list of AI models available for usage declaration include?

The list of AI models available for usage declaration should include:

1. ChatGPT: For conversational assistance, brainstorming, and drafting text.
2. GitHub Copilot: For code generation and programming support.
3. Claude: For natural language processing tasks, including summarization and content generation.
4. Grammarly: For grammar checking and writing enhancement.

5. Turnitin: For plagiarism detection and ensuring originality in writing.

6. DALL-E: For generating images and visual content related to assignments.

7. Data analysis tools (e.g., Tableau, Python libraries): For data visualization and analysis in research projects.

Inclusion of these tools will help ensure comprehensive tracking of AI usage across various academic tasks.

Should the student be given the opportunity to declare models that are not present in this list?

Yes, the system shall provide students the opportunity to declare AI models that are not present in the predefined list. This flexibility is essential to accommodate the diverse array of AI tools available and ensure that all forms of AI usage are transparently documented. Students should be able to specify the tool name, its purpose, and how it contributed to their assignment, promoting a comprehensive understanding of AI engagement in academic work.

What points should the "guided reflection" address? Should it be dependent on what the student previously declared?

The "guided reflection" should address the following points:

1. Purpose of AI Usage: Why the student chose to use specific AI tools for the assignment.

2. Impact on Learning: How the AI tools influenced their understanding and learning process.

3. Ethical Considerations: Any ethical concerns related to AI usage, including potential biases or misrepresentation of work.

4. Alignment with Academic Integrity: How their use of AI adheres to institutional policies and academic integrity standards.

5. Sustainability Practices: Any efforts made to use AI tools efficiently and sustainably.

Yes, the reflection prompts should be tailored based on the student's previous declarations to ensure relevance and encourage deeper engagement with their specific AI usage. This personalization will enhance the quality of their reflections.

Can you describe more clearly how the student should perform the "review and adjust" step basing on the feedback received?

During the "review and adjust" step, the student shall follow these actions based on the feedback received:

1. Access Feedback: The student logs into their dashboard to review the compliance feedback provided by the system regarding their AI usage.

2. Analyze Feedback: The student examines specific areas highlighted in the feedback, such as ethical concerns, sustainability practices, or adherence to academic integrity guidelines.

3. Identify Improvement Areas: Based on the feedback, the student identifies areas where their AI usage can be enhanced, such as reducing reliance on AI for certain tasks or improving their reflection on ethical implications.

4. Adjust Practices: The student modifies their approach to using AI tools in future assignments, implementing strategies to address the identified improvement areas. This may include setting personal goals for responsible AI usage or exploring additional resources on academic integrity.

5. Document Changes: The student notes these adjustments in their usage logs, ensuring a record of their commitment to continuous improvement in AI engagement.

By following these steps, the student can effectively respond to feedback and strengthen their academic practices.

What is the scale of this application? how many students are we expected to serve?

The application is designed to support a large-scale student population, ideally serving thousands of students across various academic programs. Given the increasing integration of AI tools in education, the platform should be scalable to accommodate user growth, enabling seamless access and functionality for all users while maintaining performance and compliance with institutional guidelines. Specific student capacity metrics may depend on institutional partnerships and anticipated adoption rates.

What are the guidelines for data storage and data privacy?

The guidelines for data storage and data privacy in the application shall include:

1. Data Minimization: Only collect and store data that is necessary for functionality and compliance purposes, avoiding excessive data collection.

2. Anonymization: Where possible, anonymize personal data to protect student identities and ensure privacy during data analysis and reporting.

3. Secure Storage: Implement strong encryption methods for data at rest and in transit to safeguard against unauthorized access and breaches.

4. Access Control: Limit access to personal data to authorized personnel only, ensuring that data handling complies with institutional and legal standards.

5. Compliance with Regulations: Adhere to relevant data protection regulations (e.g., GDPR, FERPA) to ensure students' rights regarding their data are respected and upheld.

6. Transparency: Provide clear information to students about what data is collected, how it is used, and their rights concerning their data.

These guidelines will help ensure that the application maintains high standards of data privacy and security while supporting responsible AI usage.

Could you clarify better the requirements about data privacy?

The data privacy requirements for the application shall include:

1. User Consent: The system shall require explicit consent from students before collecting personal data, ensuring they understand the purpose and scope of data usage.

2. Data Access Rights: The system shall provide students the ability to access, update, or delete their personal data at any time, in compliance with applicable regulations.

3. Data Retention Policy: The system shall implement a data retention policy that defines how long personal data will be stored, ensuring data is only retained as long as necessary for compliance or operational needs.

4. Incident Response Plan: The system shall have a documented incident response plan to address potential data breaches, including notification procedures for affected students.

5. Regular Audits: The system shall conduct regular audits of data handling practices to ensure compliance with privacy regulations and internal policies.

6. Privacy Training: All personnel handling personal data shall receive training on data privacy practices and institutional policies to ensure responsible data management.

These requirements ensure that the application prioritizes student privacy and complies with legal standards, fostering trust and accountability.

What are the requirements on usability?

The usability requirements for the application shall include:

1. User-Friendly Interface: The system shall provide an intuitive and accessible user interface that allows students to navigate easily and perform tasks without unnecessary complexity.

2. Responsive Design: The application shall be optimized for various devices (desktop, tablet, mobile) to ensure a consistent user experience across platforms.

3. Clear Instructions and Help Resources: The system shall offer clear instructions and help resources, including tooltips and FAQs, to assist students in understanding how to use the application effectively.

4. Feedback Mechanisms: The application shall incorporate mechanisms for students to provide feedback on usability issues, enabling continuous improvement based on user experience.

5. Minimal Learning Curve: The system shall be designed to minimize the learning curve for new users, allowing students to quickly become proficient in using the application.

6. Accessibility Compliance: The application shall adhere to accessibility standards (e.g., WCAG) to ensure that all users, including those with disabilities, can effectively use the platform.

These requirements aim to create a seamless and supportive environment for students, enhancing their engagement with AI tools and promoting responsible usage.

Are there any other users that would be using the application, apart from the students?

The primary users of the AI Guidebook are students, but other potential users may include educators and academic administrators. Educators might use the application to monitor AI usage patterns among students and provide guidance, while academic administrators could leverage the data for compliance and policy enforcement. Identifying their specific requirements will be crucial for capturing a comprehensive set of functionalities.

What are the requirements about data security?

The system shall ensure that all user data is encrypted both at rest and in transit to protect against unauthorized access. The system shall implement role-based access control to restrict access to sensitive data based on user roles. Additionally, the system shall log all access and modification activities to support auditing and compliance checks.

Other than a login page, which forms of access control should the system implement?

The system shall implement multi-factor authentication (MFA) to enhance security during user login. Additionally, the system shall provide session management controls, ensuring users are automatically logged out after a period of inactivity. Fine-grained access control should also be enforced for different features, allowing users to only access functions related to their roles.

What are the requirements about storage performance and API response time?

For storage performance, the system shall ensure data retrieval times do not exceed 200 milliseconds for logged AI usage data. Regarding API response time, the system shall provide responses to user requests within 300 milliseconds under normal operating conditions. These performance metrics are critical to maintaining a smooth user experience.

What do you mean by "normal operating conditions"?

"Normal operating conditions" refer to the typical usage scenarios of the application, where user load and system resources are within expected parameters. This encompasses standard traffic patterns, average data volume, and typical user interactions. It's important to define these conditions clearly to ensure performance metrics are achievable and realistic.

Can you define these conditions more precisely?

Normal operating conditions shall be defined as follows:

1. User Load: Up to 100 concurrent users accessing the application simultaneously.
2. Data Volume: The system shall handle an average of 10,000 logged AI usage records at any given time.
3. Interaction Type: Standard user activities include logging AI usage, querying usage patterns, and accessing ethical guidelines.

These parameters ensure that performance metrics remain realistic and achievable during typical usage scenarios.

How is the system expected to behave in abnormal conditions (for example, excessive request load)?

The system shall implement performance monitoring to identify and manage excessive request loads. It shall include load balancing mechanisms to distribute requests evenly across servers and maintain responsiveness. In case of peak load situations, the system shall provide users with feedback about potential delays and may implement rate limiting to ensure fair access. Additionally, the system shall log abnormal conditions for further analysis and improvement.

1.2 Interview Transcript Students

We interviewed a Biology student and an IT student at NTNU.

> "This is a presentation of an application. We'd like you to explain how you would interact with it. You'll first get a short description. Suppose you have access to a web application that allows you to track your AI usage in your school assignments. You can enter details about how you use AI in your everyday studying habits, and the system provides statistics and insights into your AI usage. Some of the features include: AI usage logging, assignment-level AI declarations, ethical guidelines compliance checks (according to your institution) and a personal dashboard with statistics and insights."

What kind of information would you be interested in receiving from this application?

I'd be interested in seeing how much of my actual output is influenced by AI. I'd also like to see the environmental costs of AI usage. I'm not sure how easily accessible this would be, but maybe information about which model is used and which data center performs the computation — basically where my data has been processed. I'd be interested in seeing this over time, like monthly percentages and absolute amounts.

Do you think that knowing this information — especially environmental costs and physical data location — would influence whether you use AI, which model you use, and how?

Yes, definitely. If I found out that my data was being used in places I don't support, or to an extent that I'm uncomfortable with, that would give me a lot more insight. It's very easy to just type a prompt and get an answer. But if I could see concrete numbers — for example, "this request resulted in 500 liters of water being used" — that would be much more tangible. Hard numbers would be very interesting and would definitely affect my choices. It's more meaningful than just thinking "AI is bad" in an abstract way.

The purpose of the application is not to make you use AI less, but to make you reflect on your habits and how AI affects your work. It's not meant to frame AI usage as negative, but rather to help you become more conscious of your behavior.

That makes sense. I think it's more about reflection than restriction. If I found out that my AI usage consumes a lot of water, physical space, or data storage — and that at scale, if everyone at a school did this, it would add up — then I'd want to be able to justify my AI usage. I'd want to weigh the benefits of using AI against the negative consequences. That's something the application could help with: giving insights that let users reflect and then adjust their behavior and habits based on statistics.

Do you think it could also help you evaluate the quality of your work, independently of environmental and ethical factors?

Yes. If I could see which paragraphs or sentences were directly influenced by AI, it would be easier to compare my own quality with AI-generated content. Often, when I use AI while creating something, I feel like the final product doesn't actually contain what the AI produced directly. It would be interesting to see how much of the final output is truly influenced by AI, and whether claims like "AI makes you ten times more productive" are actually true.

How do you see yourself using this tool in everyday life? Academically, or more broadly?

Mostly academically. For me, as a biology student, a lot of my studies are based on practical lab work and physical examinations, where AI isn't that helpful. AI might help with summarizing lectures or textbooks, but it presents information in a way that affects my understanding. It can feel like a prefabricated interpretation rather than my own personal understanding of the material.

So you don't see yourself using AI that much?

I'd still use it sometimes — for summaries, for example — but not heavily. And I'd be cautious, because my understanding of the course could be shaped by how AI chooses to present information.

Just to clarify: this tool isn't an AI tool itself, but a tool to log and reflect on AI usage.

In that case, I think it would be very useful. I'm not convinced that increasing AI integration into daily life is always good, so I'd like a tool that helps me see how it affects me, the environment, and the bigger picture. I'd use it to the same extent that I use AI — not necessarily a lot, but consistently. I also think it could be useful to compare AI-generated answers with my own, especially for preparing lab work, and to see where AI is insufficient versus where it's helpful.

Do you have any concerns?

Yes — how the information is collected and how much of it is shared externally. I'd want most of it to stay local.

The data would be stored locally and only used to generate personal statistics.

That's reassuring.

Are there any features you consider less important than others?

Honestly, I think they're all important. Removing one would reduce the quality of the overall assessment. Each part complements the others.

How would you like the information to be organized in the user interface?

Pie charts, bar charts, and visual comparisons — for example, AI vs non-AI content. Splitting data by courses or activities would be helpful. Ethical guidelines could be shown as a checklist. It could also be personalized: before starting, users could select what they care most about, and the dashboard could prioritize those metrics.

What would make the application easy and comfortable to use?

Good visuals — colors, fonts, clear boxes. It should be intuitive and not require too much effort.

How would you prefer to log AI usage?

As automatically as possible. Ideally, the system would detect pasted content and prompt a notification like "Log entry from AI activity," allowing me to confirm or adjust details. If it's intrusive or time-consuming, I just wouldn't use it. It should work mostly in the background, while still keeping me informed.

Are there any important topics we haven't covered?

Yes — the broader societal consequences of AI usage. Not just personal pros and cons, but impacts at scale: on education, skills, society, and climate. If data from many students could be aggregated (ethically), it could show how AI affects learning outcomes and abilities more broadly. It's complex and still under study, but very important.

That concludes our questions. Thank you very much.

1.3 Functional Requirements

From these interviews, we extracted the following requirements in the form of user stories:

1. As a student, I want to log my AI usage, so that I can track my AI habits.
2. As a student, I want to log the frequency of AI usage per each assignment, so I can see how often I rely on AI assistance.
3. As a student, I want to declare AI usage in assignments by filling out a form, so that I can ensure compliance with the ethical guidelines provided by the institution.
4. As a student, I want to refer to a personal dashboard with statistics and summaries of AI usage, so that I can get an overview of how it affects my academic everyday work.
5. As a student, I want to receive periodic personalized feedback of my AI usage, so that I can monitor my habits and compliance with the regulations.
6. As a student, I want to check my compliance with regulation at every assignment submission and at the end of every academic term, so that I can have meaningful feedback about the progress of my work.
7. As a student, I want to track my AI usage patterns over time, so that I can highlight trends in AI dependency.
8. As a student, I want to perform automated compliance checks of my AI usage against institutional guidelines, so that I can improve responsible usage aligned with academic integrity standards.
9. As a student, I want to declare the content generated by AI in my assignments, so that I can be transparent in my work.
10. As a student, I want to perform AI assignment declaration in a easy and straightforward way, so that I can do it consistently.
11. As a student, I want to specify specific tools used in each assignment and their purpose, so that I can achieve transparency in my work.
12. As a student, I want to be able to select the tools I used during assignment declaration from a list (ChatGPT, GitHub Copilot, Claude, Grammarly, Turnitin, DALL-E, data analysis tools), so that I can perform the declaration quickly.
13. As a student, I want to have the option to declare AI tools that are not present in the list during assignment declaration, so that I can feel free to use the tools that are most useful to me while still being transparent.
14. As a student, I want to receive feedback about the environmental impact of the AI usage, so that I can be more mindful and optimize that time spent with the tools.

15. As a student, I want to asses the frequency and duration of my AI tool usage, so that I can practice efficient use of AI and adhere with institutional sustainability goals.

16. As a student, I want to receive collective statistics of AI usage of the student population, so that I can better understand the societal impact of AI.

17. As a student, I want to see a chart showing the percentage breakdown of which specific AI tools (e.g., ChatGPT, Gemini, Copilot) I use most, so I can understand my preferences.

18. As a teacher, I want to use the application to monitor AI usage patterns among students, so that I can provide guidance.

19. As a teacher, I want to access student data about their AI usage, so that I can derive patters and gain insight about their study habits.

20. As a student, I want data to be organized by a personalized schedule, so that I can see the evolution of my habits and apply corrections if necessary.

21. As a student, I want the data to be categorized according to different activities by labels, so I identify areas where my work could be improved.

22. As a student, I want to access the regulatory documentation in a dedicated area of the application, so that I can assess my compliance independently on my own.

23. As a student, I want to see a pie chart showing the percentage breakdown of AI and non AI work on assignment, so I can understand my working technique.

24. As a student, I want to see a bar chart showing the percentage breakdown of each course or activity, so I can identify which courses I rely on AI for the most and adjust my study habits accordingly.

25. As an academic admin, I want the system to show an overview of the collective students' AI usage, so that I can monitor compliance for AI regulations.

26. As an academic admin, I want the system to able to group information according to different parameters, such as AI usage purpose, AI prompting and average time spent on AI, so that I can perform an analysis on the student habits.

27. As a student, I want to see which paragraphs or sentences were directly influenced by AI, so I can reflect and adjust my AI behavior.

28. As a student, I want to attach screenshots or examples of AI-generated content to my declaration, so that I can provide clear documentation.

29. As a teacher, I want to export class-level AI usage reports, so that I can share insights with department administrators.

30. As a teacher, I want to add comments or feedback on individual student AI declarations, so that I can provide personalized guidance.

31. As a teacher i want to receive an alert when a student's AI usage exceeds a predefined threshold, so that I can provide timely intervention.

32. As a student, I want to filter student AI usage data by assignment or time period, so that I can analyze specific coursework.

33. As a student, I want to see ethical guidelines as a checklist, so I can ensure that my use of AI complies with institutional regulations.

34. As a student, I want to get an overview over directly copied AI content, so I can ensure academic integrity in my assignments.

35. As a student, I want to be able to confirm and adjust details about automatically logged AI usage, so I can ensure that the detected usage reflects my actual work.

36. As a student, I want to access, update and delete my data any time, so that I control over my personal information.

37. As a system admin, I want the application to require a log in of the users, so that it can ensure security and confidentiality of personal information.

38. As a system admin, I want the system to implement multi-factor authentication, so that it can enhance security during user login.

39. As a system admin, I want the system to provide session management controls, so that it can ensure users are automatically logged out after a period of inactivity.

1.4 Non-functional Requirements

1. As a student, I want data to be clearly organized, so that I can easily derive useful information from the application.

2. As a student, I want the interface to be graphically clear, manageable and easy to use, so that I can quickly and easily use it in my everyday life.

3. As a student, I want comfortable colors, fonts and clear boxes, so that I can interact with the application with ease.

4. As a system admin, I want the system to store the data in a database, so that I can easily retrieve, update and delete information.

5. As a system admin, I want the system to implement a data retention policy that defines how long personal data will be stored, so that it can ensure that the data is only retained as long as necessary for compliance or operational needs.

6. As a system admin, I want the system to ensure data retrieval times do not exceed 200 milliseconds for logged AI usage data, so that it can ensure fast operations.

7. As a system admin, I want the system to provide responses to user requests within 300 milliseconds under normal operating conditions, so that it can maintain a smooth user experience.

8. As a system admin, I want the system to support up to 100 concurrent users accessing the application simultaneously, so that I can ensure low latency for the users.

9. As a system admin, I want the system to handle an average of 10,000 logged AI usage records at any given time. so that I can preform meaningful statistics.

10. As a system admin, I want the system to implement performance monitoring, so that I can identify and manage excessive request loads.

11. As an academic admin, I want the system to manage the data according to GDPR and other applicable data privacy regulations, so that the application is compliant with the law.

12. As a student, I want my data to be stored and used under my explicit consent, so that I can understand the scope of my data usage.

Chapter 2

Task 1.2. Requirement Review

2.1 Review Report

We reviewed the requirements document from group 04, according to the IEEE SRS quality criteria [1].

2.1.1 Correctness

> *"An SRS is correct if, and only if, every requirement stated therein is one that the software shall meet."*, paragraph 4.3.1 [1].

- Forward referencing: the requirements document does not contain forward referencing.
- Opacity: the functional requirements are clearly derived from the information contained in the interviews. However, dependencies between requirements are not very clearly specified.
- Noise: the document does not contain any useless requirements.

2.1.2 Unambiguity

> *"In SRS is unambiguous if, and only if, every requirement stated therein has only one interpretation. As a minimum, this requires that each characteristic of the final product be described using a single unique term. In cases where a term used in a particular context could have multiple meanings, the term should be included in a glossary where its meaning is made more specific."*, paragraph 4.3.2 [1]

The requirements present in the document are for the most part unambiguous. However, non-functional requirement 3 (Reliability) can have different interpretations deriving from different contexts and meanings of the phrase "the information presented".

2.1.3 Completeness

"An SRS is complete if, and only if, it includes the following elements:"

- *All significant requirements, whether relating to functionality, performance, design constraints, attributes, or external interfaces. In particular any external requirements imposed by a system specification should be acknowledged and treated.*
- *Definition of the responses of the software to all realizable classes of input data in all realizable classes of situations. Note that it is important to specify the responses to both valid and invalid input values.*
- *Full labels and references to all figures, tables, and diagrams in the SRS and definition of all terms and units of measure.*

paragraph 4.3.3 [1]

The requirements define some of the situations in which the system will be used, for example user log in, user registration, AI usage logging, AI assignment declaration and regulation compliance check. Also the behavior regarding data privacy is taken into consideration. However, the requirement list is missing any reference to the user interface, as well as specification of the behavior of the system against unexpected user behavior, invalid input or excessive load.

2.1.4 Consistency

"Consistency refers to internal consistency. If an SRS does not agree with some higher-level document, such as a system requirements specification, then it is not correct.", paragraph 4.3.4 [1]

An SRS is consistent if and only if no subset of individual requirements is in conflict. The requirements present in the document contain no conflicts, as each one specifies a different feature that is independent on the others.

2.1.5 Ranking

"An SRS is ranked for importance and/or stability if each requirement in it has an identifier to indicate either the importance or stability of that particular requirement.", paragraph 4.3.5 [1]

The requirements are not ranked, though there is a reference to the future hiearchy of importance they will be structured with.

2.1.6 Verifiability

"An SRS is verifiable if, and only if, every requirement stated therein is verifiable. A requirement is verifiable if, and only if, there exists some finite cost-effective process with which a person or machine can check that the software product meets the requirement. In general any ambiguous requirement is not verifiable.", paragraph 4.3.6 [1]

The functional requirements, although not very precisely specified, are loosely verifiable. The non-functional requirements 1 and 2 can be tested more exactly due to their measurable formulation. Non functional requirements 3 and 4 are not testable, as they are not stated in a measurable way.

2.1.7 Modifiability

> *"An SRS is modifiable if, and only if, its structure and style are such that any changes to the requirements can be made easily, completely, and consistently while retaining the structure and style."*, paragraph 4.3.7 [1]

Due to the absence of dependencies between the different requirements, all the requirements appear modifiable. There is no redundancy (all requirements appear in only one place) and all requirements are expressed separately.

2.1.8 Traceability

> *"An SRS is traceable if the origin of each of its requirements is clear and if it facilitates the referencing of each requirement in future development or enhancement documentation. The following two types of traceability are recommended:"*
>
> - *Backward traceability (i.e., to previous stages of development). This depends upon each requirement explicitly referencing its source in earlier documents.*
> - *Forward traceability (i.e., to all documents spawned by the SRS). This depends upon each requirement in the SRS having a unique name or reference number.*
>
> paragraph 4.3.8 [1]

Non-functional requirements 2 and 4 are not backward-traceable, as there is no reference to availability and scalability requirements in the two conducted interviews. As for forward traceability, the requirements are not clearly labeled.

2.2 Revised Requirement Document

This is the changed and updated requirements based on feedback received both from Group 4 during the peer-review (see Appendix) and from the teaching staff (on https://innspill.ai/no/).

One improvement involved restructuring the requirement list. Related requirements were grouped together to improve readability. This was proposed by Group 4, who wrote "20 and 21, what data? The one in the dashboard? Would put these closer together to give better context." in the peer-review feedback. While requirements should remain logically independent, reorganizing them makes the document easier to navigate.

2.2.1 Core logging requirement

The following feedback regarding the logging-related requirements was provided by the teaching staff.

> "Logging + frequency + duration
>
> FR1 (log usage) overlaps heavily with FR2 (log frequency), FR7 (track patterns over time), FR15 (assess frequency/duration), FR20 (organized by schedule), FR32 (filter by time period).
>
> These should be one core logging requirement + a few analytics/dashboard refinements."

Based on this, the overlapping requirements was reviewed. FR7 was general and could be merged with FR1 and FR2 into a single requirement. FR1 contained the core phrase, "... so that i can track my AI habits", which was consistently emphasized in the iRE interview. Since the only meaningful way to log AI usage is per assignment, the functionality from FR2 was incorporated into the new revised requirement. The revised core logging requirement is:

"As a student, I want to log my AI usage on each assignment, so that I can track my AI habits."

FR15 was kept due to its relevance to sustainability. FR20 and FR32 were removed, as their functionality was overlapping and redundant.

2.2.2 Declaration workflow

The teaching staff provided the following feedback regarding the requirements related to the declaration workflow.

> "Declaration workflow
>
> FR3 (declare in form), FR9 (declare content generated by AI), FR10 (easy declaration), FR11 (specify tools/purpose), FR12 (select tools from list), FR13 (declare tools not in list), FR28 (attach screenshots/examples). These can be reorganized into a clean chain: Declare → add tool details → tool list + "other" option → attach evidence (optional)."

In response, the requirements was reorganized into a clearer, more structured workflow. The revised order follows the chain suggested in the feedback. Starting with the declaration form (FR3), then specifying tool details (FR11), selecting tools from a predefined list (FR12), adding tools not included in the list (FR13), declaring which content was generated by AI (FR9), and finally attaching optional evidence (FR28). Grouping these together provides a more logical flow while keeping individual requirements independent.

FR10 was moved to the list of non-functional requirements, as described in the "NFR quality issues" (see Subsection 2.2.6), where it was merged appropriately.

2.2.3 Dashboards/charts

> "FR4 (dashboard overview) overlaps with FR17 (tool breakdown chart), FR23 (AI vs non-AI pie chart), FR24 (bar chart per course/activity). These are refinements of one "dashboard visualization" capability."

FR4 represents the general dashboard overview and is therefore kept unchanged. It also clarifies that the dashboard contains multiple charts rather than a single visualization. To address the overlap and improve consistency, FR17, FR23, and FR24 were refined into more precise requirements:

"As a student, I want to see the percentage breakdown of which specific AI tools I use most in form of a charts on my personal dashboard, so I can understand my preferences."
"As a student, I want to see the percentage breakdown of AI and non AI work on assignment as a chart on my personal dashboard, so I can understand my working technique."
"As a student, I want to see the percentage breakdown of each course or activity as a chart on my personal dashboard, so I can identify which courses I rely on AI for the most and adjust my study habits accordingly."

2.2.4 Compliance checking

> "FR6 (check compliance at submission & end of term) overlaps with FR8 (automated compliance checks), FR5 (personalized feedback), FR34 (overview of copied content). Needs clearer distinction."

After reviewing, FR6 and FR5 were found to be duplicates, and therefore FR5 was removed. FR34 originated as a user wish expressed during the student interviews. As mentioned in the exercise lecture (on Friday 6/2/2026), wishes should not be included in the formal requirement list, and it was therefore also removed.

FR6 and FR8 were merged to form a more general requirement that captures both automated compliance checking and the timing of feedback. The revised requirement is:

"As a student, I want to receive automatic feedback on my compliance with regulations at every assignment submission and at the end of every academic term, so that I can improve responsible usage aligned with academic integrity standards"

2.2.5 Additional modifications

Based on the feedback, some additional requirements were adjusted for consistency and to reduce redundancy. FR22 and FR33 were found to overlap, and therefore they were merged into a single requirement. This requirement now appears as the new FR17 in the updated requirements list (see Section 2.3).

"As a student, I want to access the regulatory documentation as a checklist in a dedicated area of the application, so that I can assess my compliance independently on my own"

2.2.6 NFR quality issues

> "Many NFRs are phrased as "I want comfortable colors" (NFR3) which is not measurable. Some are good (latency targets, concurrent users), but they need consistent structure."

NFR1, NFR2, NFR3, and FR10 were identified as non-measurable and therefore required revision. Since they all addressed similar aspects of usability, they were merged into a single requirement measured by the System Usability Scale [2]:

"As a student, I want the interface to score an average above 80 on the System Usability Scale (SUS), so that I can quickly and easily use it in my everyday life."

Additionally, the peer-review feedback from Group 4 noted that NFR7 lacked a clear definition of normal operating conditions. These definitions is defined as NFR2 and NFR3 in the updated non-functional requirements list (see Section 2.4). This has now been addressed by explicitly specifying these conditions within the revised requirement:

"As a system admin, I want the system to provide responses to user requests within 300 milliseconds under normal operating conditions (defined as a load of up to 100 concurrent users and 10 000 logged AI usage records), so that it can maintain a smooth user experience."

2.3 Revised Functional Requirements

After feedback from the teaching staff and the peer review conducted by Group 4, the functional requirements have been revised as follows.

1. As a student, I want to log my AI usage on each assignment, so that I can track my AI habits.
2. As a student, I want to asses the frequency and duration of my AI tool usage, so that I can practice efficient use of AI and adhere with institutional sustainability goals.
3. As a student, I want to declare AI usage in assignments by filling out a form, so that I can ensure compliance with the ethical guidelines provided by the institution.
4. As a student, I want to specify specific tools used in each assignment and their purpose, so that I can achieve transparency in my work.
5. As a student, I want to be able to select the tools I used during assignment declaration from a list (ChatGPT, GitHub Copilot, Claude, Grammarly, Turnitin, DALL-E, data analysis tools), so that I can perform the declaration quickly.
6. As a student, I want to have the option to declare AI tools that are not present in the list during assignment declaration, so that I can feel free to use the tools that are most useful to me while still being transparent.

7. As a student, I want to declare the content generated by AI in my assignments, so that I can be transparent in my work.

8. As a student, I want to attach screenshots or examples of AI-generated content to my declaration, so that I can provide clear documentation.

9. As a student, I want to refer to a personal dashboard with statistics and summaries of AI usage, so that I can get an overview of how it affects my academic everyday work.

10. As a student, I want to see the percentage breakdown of which specific AI tools I use most in form of a charts on my personal dashboard, so I can understand my preferences.

11. As a student, I want to see the percentage breakdown of AI and non AI work on assignment as a chart on my personal dashboard, so I can understand my working technique.

12. As a student, I want to see the percentage breakdown of each course or activity as a chart on my personal dashboard, so I can identify which courses I rely on AI for the most and adjust my study habits accordingly.

13. As a student, I want to receive automatic feedback on my compliance with regulations at every assignment submission and at the end of every academic term, so that I can improve responsible usage aligned with academic integrity standards

14. As a student, I want to access the regulatory documentation as a checklist in a dedicated area of the application, so that I can assess my compliance independently on my own

15. As a student, I want to receive feedback about the environmental impact of the AI usage, so that I can be more mindful and optimize that time spent with the tools.

16. As a student, I want to receive collective statistics of AI usage of the student population, so that I can better understand the societal impact of AI.

17. As a student, I want the data to be categorized according to different activities by labels, so I identify areas where my work could be improved.

18. As a student, I want to be able to confirm and adjust details about automatically logged AI usage, so I can ensure that the detected usage reflects my actual work.

19. As a student, I want to access, update and delete my data any time, so that I control over my personal information.

20. As a teacher, I want to use the application to monitor AI usage patterns among students, so that I can provide guidance.

21. As a teacher, I want to access student data about their AI usage, so that I can derive patters and gain insight about their study habits.

22. As a teacher, I want to export class-level AI usage reports, so that I can share insights with department administrators.

23. As a teacher, I want to add comments or feedback on individual student AI declarations, so that I can provide personalized guidance.

24. As a teacher i want to receive an alert when a student's AI usage exceeds a predefined threshold, so that I can provide timely intervention.

25. As an academic admin, I want the system to show an overview of the collective students' AI usage, so that I can monitor compliance for AI regulations.

26. As an academic admin, I want the system to able to group information according to different parameters, such as AI usage purpose, AI prompting and average time spent on AI, so that I can perform an analysis on the student habits.

27. As a system admin, I want the application to require a log in of the users, so that it can ensure security and confidentiality of personal information.

28. As a system admin, I want the system to implement multi-factor authentication, so that it can enhance security during user login.

29. As a system admin, I want the system to provide session management controls, so that it can ensure users are automatically logged out after a period of inactivity.

2.4 Revised Non-functional Requirements

Similarly as the functional requirements, the non-functional requirements have been updated based on the same feedback and are presented below.

1. As a student, I want the interface to score an average above 80 on the System Usability Scale (SUS), so that I can quickly and easily use it in my everyday life.

2. As a system admin, I want the system to support up to 100 concurrent users accessing the application simultaneously, so that I can ensure low latency for the users.

3. As a system admin, I want the system to handle an average of 10,000 logged AI usage records at any given time. so that I can preform meaningful statistics.

4. As a system admin, I want the system to provide responses to user requests within 300 milliseconds under normal operating conditions (defined as a load of up to 100 concurrent users and 10 000 logged AI usage records), so that it can maintain a smooth user experience.

5. As a system admin, I want the system to store the data in a database, so that I can easily retrieve, update and delete information.

6. As a system admin, I want the system to implement a data retention policy that defines how long personal data will be stored, so that it can ensure that the data is only retained as long as necessary for compliance or operational needs.

7. As a system admin, I want the system to ensure data retrieval times do not exceed 200 milliseconds for logged AI usage data, so that it can ensure fast operations.

8. As a system admin, I want the system to implement performance monitoring, so that I can identify and manage excessive request loads.

9. As an academic admin, I want the system to manage the data according to GDPR and other applicable data privacy regulations, so that the application is compliant with the law.

10. As a student, I want my data to be stored and used under my explicit consent, so that I can understand the scope of my data usage.

Chapter 3

Task 1.3. Requirement dependency analysis and categorization

The dependency structure has a clear impact on the implementation order by showing which requirements are foundational and which depend on others. Core functional requirements such as FR1, FR3, and FR4 enable a large number of other requirements and should therefore be implemented early to avoid blocking later development. These requirements form the basis for many dependent features, making them critical for both system functionality and architectural stability. Several non-functional requirements, especially NFR4, act as constraints across multiple functional requirements, which increases implementation risk if they are delayed or changed late in the process.

The presence of "enables" and "refines" relationships supports an incremental implementation strategy. Basic functionality can be developed first, while more detailed or specialized requirements are added or refined in later iterations. This reduces risk and supports better change handling, since refinements build on already implemented behavior rather than requiring redesign. Dependencies between functional and non-functional requirements also show that certain quality attributes, such as security and performance, must be considered early to ensure consistency and testability across the system.

From a planning perspective, highly connected requirements require more careful scheduling, estimation, and testing due to their broad impact. Less connected or optional requirements can be implemented later with lower risk. Overlapping requirements highlight areas where coordination is necessary to avoid redundancy and maintain modifiability. Overall, understanding these dependencies supports better sequencing, reduces rework, enables effective change impact analysis, and helps manage technical and integration risks throughout the development process.

3.1 Clustering

We have identified the following clustering of requirements:

- C1 - Student data capture and logging: FR1, FR2, FR18, NFR5
- C2 - Assignment declaration: FR3, FR4, FR5, FR6, FR7, FR8, FR13

- C3 - Dashboard and UI: FR9, FR10, FR11, FR12, FR13, FR14, FR15, FR16, FR17, NFR1
- C4 - Compliance and feedback: FR13, FR14, FR15, FR16, FR23
- C5 - Teacher/admin reporting: FR20, FR21, FR22, FR23, FR24, FR25, FR26
- C6 - Security and privacy: FR19, FR27, FR28, FR29, NFR6, NFR9, NFR10
- C7 - Performance: NFR2, NFR3, NFR4, NFR7, NFR8

3.2 JSON dependency file

The following JSON definition file reports the identified relationships among the requirements, as well as their categorization and ranking (critical/optional/derived).


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"FR1": {
  "depends_on": [],
  "enables": ["FR2", "FR3", "FR9"],
  "conflicts_with": [],
  "refines": [],
  "duplicates": [],
  "overlaps": ["FR3"],
  "category": ["C1"],
  "ranking": ["critical"]
},
"FR2": {
  "depends_on": ["FR1", "NFR5"],
  "enables": [],
  "conflicts_with": [],
  "refines": [],
  "duplicates": [],
  "overlaps": [],
  "category": ["C1"],
  "ranking": ["derived"]
},
"FR3": {
  "depends_on": ["FR1"],
  "enables": ["FR4", "FR7"],
  "conflicts_with": [],
  "refines": ["FR1"],
  "duplicates": [],
  "overlaps": ["FR1"],
  "category": ["C2"],
  "ranking": ["critical"]
},
```

```
"FR4": {
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  "enables": ["FR5"],
  "conflicts_with": [],
  "refines": [],
  "duplicates": [],
  "overlaps": [],
  "category": ["C2"],
  "ranking": ["critical"]
},
"FR5": {
  "depends_on": ["FR4"],
  "enables": ["FR6"],
  "conflicts_with": [],
  "refines": ["FR4"],
  "duplicates": [],
  "overlaps": [],
  "category": ["C2"],
  "ranking": ["critical"]
},
"FR6": {
  "depends_on": ["FR5"],
  "enables": [],
  "conflicts_with": [],
  "refines": ["FR4"],
  "duplicates": [],
  "overlaps": [],
  "category": ["C2"],
  "ranking": ["derived"]
},
"FR7": {
  "depends_on": ["FR3"],
  "enables": ["FR8"],
  "conflicts_with": [],
  "refines": [],
  "duplicates": [],
  "overlaps": [],
  "category": ["C2"],
  "ranking": ["critical"]
},
"FR8": {
  "depends_on": ["FR7"],
  "enables": [],
  "conflicts_with": [],
  "refines": ["FR7"],
  "duplicates": [],
```

```
  "overlaps": [],
  "category": ["C2"],
  "ranking": ["critical"]
},
"FR9": {
  "depends_on": ["NFR5"],
  "enables": [],
  "conflicts_with": [],
  "refines": [],
  "duplicates": [],
  "overlaps": [],
  "category": ["C3"],
  "ranking": ["critical"]
},
"FR10": {
  "depends_on": ["FR9"],
  "enables": [],
  "conflicts_with": [],
  "refines": ["FR9"],
  "duplicates": [],
  "overlaps": [],
  "category": ["C3"],
  "ranking": ["optional"]
},
"FR11": {
  "depends_on": ["FR9"],
  "enables": [],
  "conflicts_with": [],
  "refines": ["FR9"],
  "duplicates": [],
  "overlaps": [],
  "category": ["C3"],
  "ranking": ["optional"]
},
"FR12": {
  "depends_on": ["FR9"],
  "enables": [],
  "conflicts_with": [],
  "refines": ["FR9"],
  "duplicates": [],
  "overlaps": [],
  "category": ["C3"],
  "ranking": ["optional"]
},
"FR13": {
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```

```
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  "refines": [],
  "duplicates": [],
  "overlaps": [],
  "category": ["C2", "C3", "C4"],
  "ranking": ["critical"]
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"FR14": {
  "depends_on": ["FR13"],
  "enables": [],
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  "refines": [],
  "duplicates": [],
  "overlaps": [],
  "category": ["C3", "C4"],
  "ranking": ["derived"]
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"FR15": {
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  "enables": [],
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  "refines": [],
  "duplicates": [],
  "overlaps": [],
  "category": ["C3", "C4"],
  "ranking": ["optional"]
},
"FR16": {
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  "enables": [],
  "conflicts_with": [],
  "refines": [],
  "duplicates": [],
  "overlaps": [],
  "category": ["C3", "C4"],
  "ranking": ["optional"]
},
"FR17": {
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  "enables": [],
  "conflicts_with": [],
  "refines": [],
  "duplicates": [],
  "overlaps": [],
  "category": ["C3"],
```

```
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  "refines": [],
  "duplicates": [],
  "overlaps": [],
  "category": ["C1"],
  "ranking": ["critical"]
},
"FR19": {
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  "enables": [],
  "conflicts_with": [],
  "refines": [],
  "duplicates": [],
  "overlaps": [],
  "category": ["C6"],
  "ranking": ["critical"]
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  "ranking": ["critical"]
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  "enables": ["FR20", "FR22", "FR23", "FR24"],
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  "refines": [],
  "duplicates": [],
  "overlaps": [],
  "category": ["C5"],
  "ranking": ["critical"]
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"FR22": {
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  "enables": [],
  "conflicts_with": [],
```

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  "overlaps": [],
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  "ranking": ["critical"]
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  "enables": [],
  "conflicts_with": [],
  "refines": [],
  "duplicates": [],
  "overlaps": [],
  "category": ["C4", "C5"],
  "ranking": ["critical"]
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  "enables": [],
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  "refines": [],
  "duplicates": [],
  "overlaps": [],
  "category": ["C5"],
  "ranking": ["critical"]
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"FR25": {
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  "enables": [],
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  "refines": [],
  "duplicates": [],
  "overlaps": [],
  "category": ["C5"],
  "ranking": ["critical"]
},
"FR26": {
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  "enables": [],
  "conflicts_with": [],
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  "ranking": ["critical"]
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```

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  "ranking": ["critical"]
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"FR28": {
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  "ranking": ["critical"]
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  "refines": [],
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  "overlaps": [],
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  "ranking": ["critical"]
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  "conflicts_with": [],
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  "ranking": ["critical"]
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  "refines": [],
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  "ranking": ["optional"]
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  "ranking": ["critical"]
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"NFR5": {
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  "ranking": ["critical"]
},
"NFR7": {
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  },
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    "overlaps": [],
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    "ranking": ["critical"]
  },
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    "refines": [],
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    "overlaps": [],
    "category": ["C6"],
    "ranking": ["critical"]
  },
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    "enables": [],
    "conflicts_with": [],
    "refines": [],
    "duplicates": [],
    "overlaps": [],
    "category": ["C6"],
    "ranking": ["critical"]
  }
}
```

Requirements Review (Group 3)

General things

- Some non-functional requirements should probably be functional. Login and two-factor authentication should either be functional requirements, as a system cannot require that users log in if it does not support the functionality to register and authenticate users.
- We would recommend grouping relevant user stories, such as those concerning a personalised dashboard, and its functionality being together.
- There are a lot of requirements that repeat the same premise of previously stated requirements just worded differently or with a small additive change.
    - Functional requirements (4, 7 and 15) all discuss AI usage tracking. Could be shortened to one functional requirement.

Interview Review Feedback

- Some questions can be a bit leading. This one implies that it could “help” for example
    - *“Do you think it could also help you evaluate the quality of your work, independently of environmental and ethical factors?”*
- Could instead be written like:
    - *“What factors are important to consider when evaluating the quality of your work?”*
- Great work going in-depth on the test subject’s thoughts by asking follow up questions!

Role Explanation (Optional : Not important)

- In non-functional requirements (6 - 7) there are users like “academic admin” and “system admin”. These user’s purpose can be ambiguous if their role is not properly explained beforehand. Should include a role explanation for the included users in functional and non-functional requirements. The academic admin is mentioned in the iRE agent interview, but their role is never defined.

User stories - Feedback

- User stories follow the structured user stories format which is great!
- Short, detailed and concise user stories.
- Lacks ranking (optional, essential etc.,)
- Clear separation of different users

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